Raj Ojha
LinkedIn — GitHub
Email : raj.2428cseai1369@kiet.edu
Mobile : 9336985987

Education

- **KIET Group of Institutions** Ghaziabad, Uttar Pradesh
  B.Tech in Computer Science & Engineering (Artificial Intelligence), CGPA: 9.02 *2024 – Present*

Projects

- **Autonomous Spacecraft Docking using Reinforcement Learning**: Built a real-time 3D spacecraft docking simulation using Unity and Python to reduce human intervention. Implemented custom physics and collision handling for multi-object (two-spacecraft) interactions. Trained autonomous control policies using DQN and PPO with reward functions based on distance minimization and collision penalties. Training was performed on NVIDIA A100 GPU infrastructure.
- **Real-time Collaborative Coding Platform**: Developed a real-time multi-user collaborative coding platform using React, Python, and Socket.IO. Implemented live code synchronization, cursor presence, auto-scroll, syntax highlighting, and support for multiple programming languages. Designed conflict-resolution logic to maintain consistent shared state across concurrent users.
- **Official College Application Website (ICICI 2026 Portal)**: Designed and deployed a production-grade full-stack web application used by students for official college applications. Built the frontend using React and backend REST APIs using Node.js, with complete form validation, workflow handling, and database integration. The system is live and actively used.
- **Handwritten Digit Recognition using CNN**: Implemented an end-to-end handwritten digit recognition system using a Convolutional Neural Network trained on the MNIST dataset. Performed custom data preprocessing and trained the model using Keras (TensorFlow backend), achieving approximately 93% accuracy.

Skills

- **Programming Languages:** C/C++, Java, Python, JavaScript, TypeScript
- **Data Structures & Algorithms:** Intermediate
- **Web & Backend:** React, Node.js, REST APIs, WebSockets, MongoDB, SQL, Vite
- **Machine Learning & AI:** Machine Learning, Deep Learning (CNNs), Reinforcement Learning (DQN, PPO), Computer Vision, Keras, TensorFlow
- **Tools & Systems:** Git, GitHub, Linux, Docker, AWS (Foundational Certification)

Achievements & Hackathons

- **Winner – CybersecureX (National Level Hackathon)**: Secured 1st position in a national-level cybersecurity hackathon focused on identifying, exploiting, and mitigating real-world security vulnerabilities. Demonstrated skills in threat analysis, secure system design, and defensive security strategies.
- **Winner – SIG+H Hackathon (Institute Level)**: Won the institute-level SIG+H hackathon by developing a technical solution addressing cybersecurity and system reliability challenges.
- **Academic Performance**: Maintained a CGPA of 9.02 in B.Tech CSE (Artificial Intelligence).